

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 1: Short Answer Test**

**COURSE NAME:** Cloud Computing and Big Data Analytics      **COURSE CODE:** CSA15

---

**COURSE OUTCOME COVERED**

**CO1:** Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)  
Question Count: 5

**Duration: 60 Minutes**

**Total Marks: 25**

---

**Code of Conduct**

[192321138]

I Ganeshkardick M (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ganeshkardick M

| Criteria  | Conceptual Accuracy (1.5) | Coverage of Concepts (1) | Use of Examples (0.75) | Comparison / Differentiation (1) | Clarity of Expression (0.75) | TOTAL (5) |
|-----------|---------------------------|--------------------------|------------------------|----------------------------------|------------------------------|-----------|
| Weightage | 30%                       | 20%                      | 15%                    | 20%                              | 15%                          | 100%      |
| Q1        |                           |                          |                        |                                  |                              |           |
| Q2        |                           |                          |                        |                                  |                              |           |
| Q3        |                           |                          |                        |                                  |                              |           |
| Q4        |                           |                          |                        |                                  |                              |           |
| Q5        |                           |                          |                        |                                  |                              |           |

**Signature of the Faculty**

**Total Marks Awarded:**

## Assessment Tasks

### Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

### Short Answer Test Rubric

| Criteria                     | Weightage | Level 4 - Exemplary                                           | Level 3 - Proficient                          | Level 2 - Developing                         | Level 1 - Beginning                 |
|------------------------------|-----------|---------------------------------------------------------------|-----------------------------------------------|----------------------------------------------|-------------------------------------|
| Conceptual Accuracy          | 30%       | Accurate definitions and precise explanation of all key terms | Mostly accurate with minor gaps               | Partial understanding with noticeable errors | Incorrect or irrelevant explanation |
| Coverage of Concepts         | 20%       | Covers all expected sub-points completely                     | Covers most expected points                   | Covers only basic points                     | Very limited coverage               |
| Use of Examples              | 15%       | Uses highly relevant cloud examples to strengthen the answer  | Uses relevant examples with minor mismatch    | Uses vague or partially relevant examples    | No useful examples                  |
| Comparison / Differentiation | 20%       | Clearly distinguishes concepts with strong logic              | Reasonable differentiation with minor overlap | Comparison is weak or incomplete             | Fails to differentiate concepts     |
| Clarity of Expression        | 15%       | Clear, organized, and concise answer writing                  | Generally clear with few issues               | Understandable but poorly organized          | Difficult to follow                 |

## ASSESSMENT TOOL - 1

### CLOUD COMPUTING - CSA1516

Ganesh Karthick M  
[192321138]

B.Tech.IT.

1. Describe the evolution from standalone computing to cloud-based systems. Summarise key technologies / traditions that enabled cloud computing.

- \* Standalone computers worked independently with local storage.

- \* Then networks connected computers to store data.

- \* The internet made online communication possible.

- \* Virtualization allowed multiple virtual users to access data and application over internet.

2. Interpret the working of hypervisors in virtualization differentiate between Type 1 and Type 2 hypervisors with examples.

A hypervisor is a software that creates and manages virtual machines.

| Type 1 hypervisor                                                                                                                              | Type 2 hypervisor                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>* Runs directly on hardware</li> <li>* Easier and more secure</li> <li>* used in data center</li> </ul> | <ul style="list-style-type: none"> <li>* Runs on host OS</li> <li>* Easier to install and use.</li> <li>* used on Personal Computers.</li> </ul> |

3. Describe the role of API's in cloud service interoperation, explain how API's support interoperability between services.

An API (Application Programming Interface) is set of rules that allows different cloud applications and services to communicate with each other.

API help integrate cloud service by enabling the exchange of data and functions without requiring users to know the internal details of the system. more function - efficient and easy to use.

Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.

\* Resource pooling means sharing computing resources like servers and storage among many users.

\* Resource cost improves efficiently and supports many users at the same time.

\* Supports multiple users at the same time without affecting performance.

---

5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Rapid elasticity means cloud resources can increase or decrease automatically based on demands.

## How it supports dynamic Workload.

- \* Resources can be scaled up during high demand.

- \* Resources can be scaled down when demand decreases.

- \* Avoids resource wastage.

- \* Reduces operational wastage.

- \* provides flexibility to handle changing overloads

eg: During an online shopping sale cloud resources automatically increase to handle more users and reduce after the sale ends.

---